

A Living Task-Transfer Graph for Compounding Auto-Research

Pilot Findings and Full-Scale Outlook

Brain medical-imaging domain · 886-paper subsample · 2026-06-01 · github.com/vveii14/Auto-Research

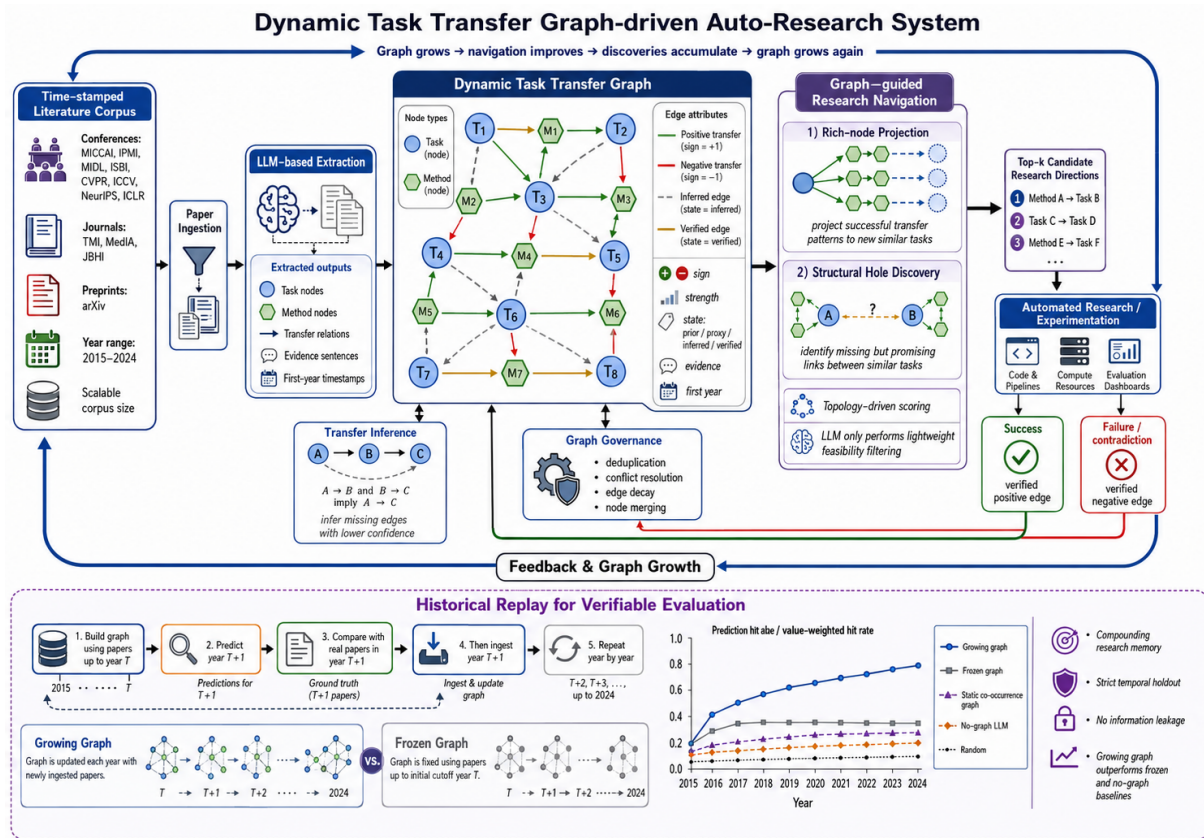


Figure 0. System overview: corpus → LLM extraction (task/method nodes + directed transfer edges) → navigation (rich-node projection + structural holes) → explore-and-fold-back (graph grows) → governance. Bottom: the history-as-label temporal-replay evaluation.

1. Executive Summary

We cast automatic research-direction discovery as **candidate transfer-edge prediction on a temporal graph**. From timestamped papers we extract task and method nodes together with **directed, signed “does A help B” transfer edges**, forming a graph that grows as exploration proceeds and, in turn, navigates it; a **history-as-label** temporal replay tests whether it becomes more accurate over time. This report presents an end-to-end pilot on 886 brain medical-imaging abstracts.

Key findings (stated honestly):

- **Significant.** Graph-structured navigation predicts future research directions **significantly better than a no-graph LLM, a co-occurrence graph, and random** (paired bootstrap CI excludes 0). This also shows the gain comes from graph structure, not from the LLM memorizing future papers.
- **Not yet significant.** The headline **compounding** effect (growing > frozen) is consistently positive in the point estimate, but its CI still spans 0 at the 900-paper scale.
- **Real weakness.** 78% of method->task edges are mere *usage* rather than genuine *transfer*; only ~23% of all edges are genuine transfer. The main cause is the extraction prompt (fixable).
- **Verdict.** The infrastructure and the “graph > baselines” result are established (a safety-net contribution); the compounding claim warrants one scale-up with a clear stop-loss.

2. Core Idea and Workflow

Positioning. The first system to turn a research-relation graph from a *static, read-only substrate* into a structure that **grows with exploration and navigates it**. Closest prior work: Intern-Atlas (static, read-only), Deep Ideation (fixed substrate), AC/DC (no relational graph). The novelty is the **combination of directed transfer semantics + living growth + graph navigation + compounding**, applied equally to method->task and task->task edges.

Workflow (8 stages):

- Define corpus scope (a tunable parameter: venues / years / sample size).
- Ingest timestamped papers indiscriminately.
- LLM extraction: task/method nodes + directed transfer edges, each with a verbatim evidence span.
- Transitive inference: A->B and B->C imply A->C, filling sparse regions.
- Navigation (non-random, the core): rich-node projection + structural holes + topological scoring; the LLM only performs caged local filtering, yielding top-k directions.
- Explore and fold back: a direction is sent to auto-research; success and failure alike return as verified edges, so the graph grows.
- Governance: deduplication / conflict resolution / decay, to prevent the graph from degrading.
- Continuous ingestion of new papers, so the graph grows with the field.

Evaluation. Build the graph up to year T, predict the transfer edges that actually emerge in T+1, score first and only then ingest T+1, and roll forward year by year; compare growing / frozen / co-occurrence / no-graph-LLM / random. Real publication history provides **free, objective labels**, which avoids circular reasoning.

3. Pilot Setup and Corpus

Source: Semantic Scholar (abstracts only); filtered to fieldsOfStudy = Computer Science plus brain-imaging / method terms; years 2015-2024. 8,000 papers were fetched; the pilot uses an 886-paper subsample (500 from ≤ 2020 to build the graph, and 100 per year for 2021-2024 as prediction targets / ingestion material). Extraction uses an Azure-hosted Opus model; node embeddings use all-MiniLM-L6-v2.

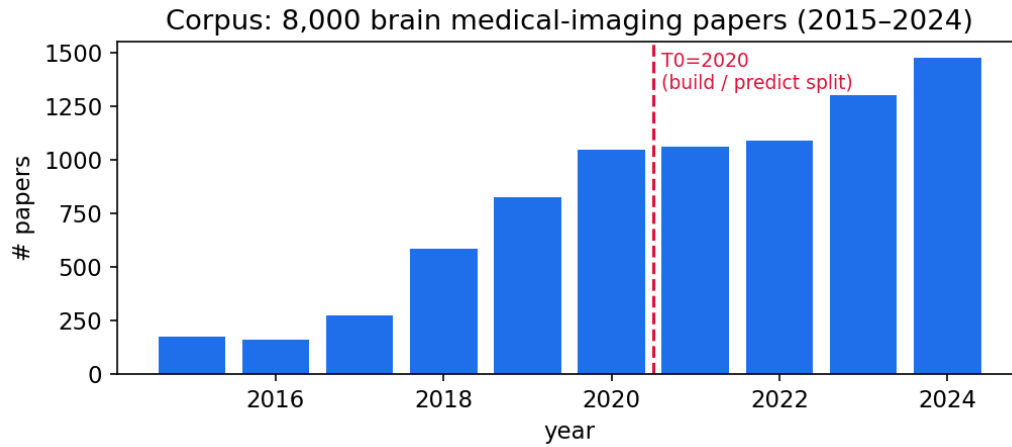


Figure 1. Corpus year distribution. The steady 2015-2024 growth suits temporal hold-out; the red line marks the T0=2020 build/predict split.

4. The Constructed Transfer Graph

886 papers yield, after entity resolution, **2,907 nodes (1,064 task / 1,843 method)** and **1,748 extracted edges**; transitive inference adds about 518 more. Edges are dominated by method->task (1,581); task->task accounts for 131 and method->method for 36.

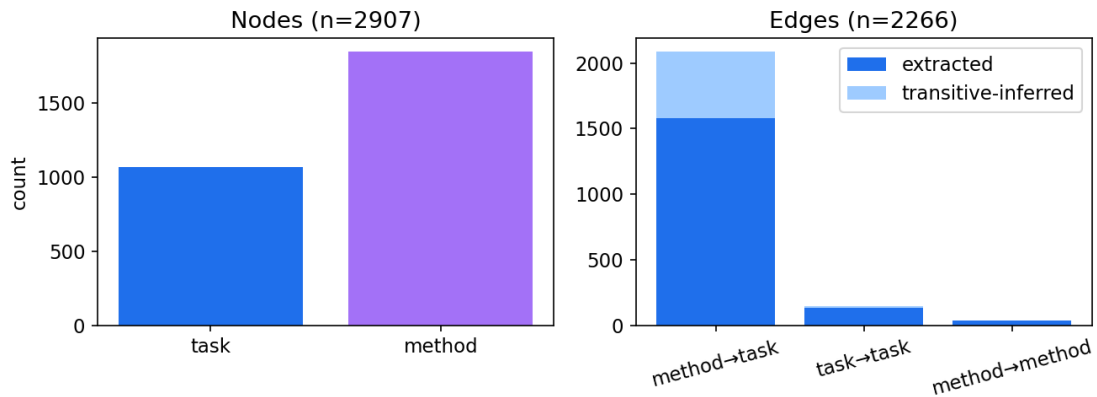


Figure 2. Graph composition: node types (left) and edge types (right, stacked to show extracted vs. transitively inferred).

Representative genuine task->task transfer edges (who-helps-whom):

Source task -> Target task	Evidence (excerpt)
age prediction (structural MRI) -> Alzheimer's diagnosis	sensitive to deviance from normal aging...
hippocampus segmentation -> Alzheimer's classification	multi-task CNN jointly learning hippocampus...
virtual connectome inference -> AD classification	trained on virtual connectomes can be used...
PET segmentation <-> denoising <-> partial-volume corr.	segmentation can help in denoising and PVC...

5. Pilot Results

5.1 Navigation accuracy (semantic matching, bootstrap 95% CI)

On the same temporal-forecast task, our arms (growing / frozen) consistently outperform the co-occurrence graph, the no-graph LLM, and random across all k. The no-graph LLM is near-random, indicating that the LLM alone (with its memorized future) is weak and that the gain comes from the graph.

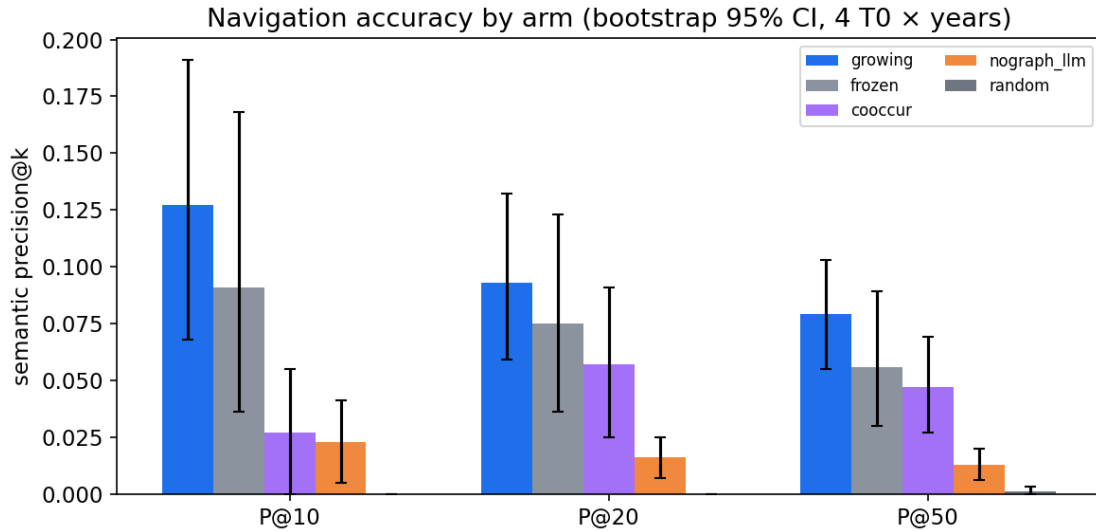


Figure 3. Semantic precision@k by arm with 95% confidence intervals (pooled over 4 T0 settings x years).

Arm	P@10	P@20	P@50
growing	0.127	0.093	0.079
frozen	0.091	0.075	0.056
co-occurrence	0.027	0.057	0.047
no-graph LLM	0.023	0.016	0.013
random	0.000	0.000	0.001

5.2 Compounding probe: growing vs. frozen

Sweeping the start year T0: when the base is small and growth is proportionally large (T0=2017, 2019), growing clearly exceeds frozen; when the base is large and yearly increments are tiny (T0=2020), they converge. This is consistent with “growth must be large relative to the base for compounding to surface,” and reflects a limitation of the small subsample at late T0.

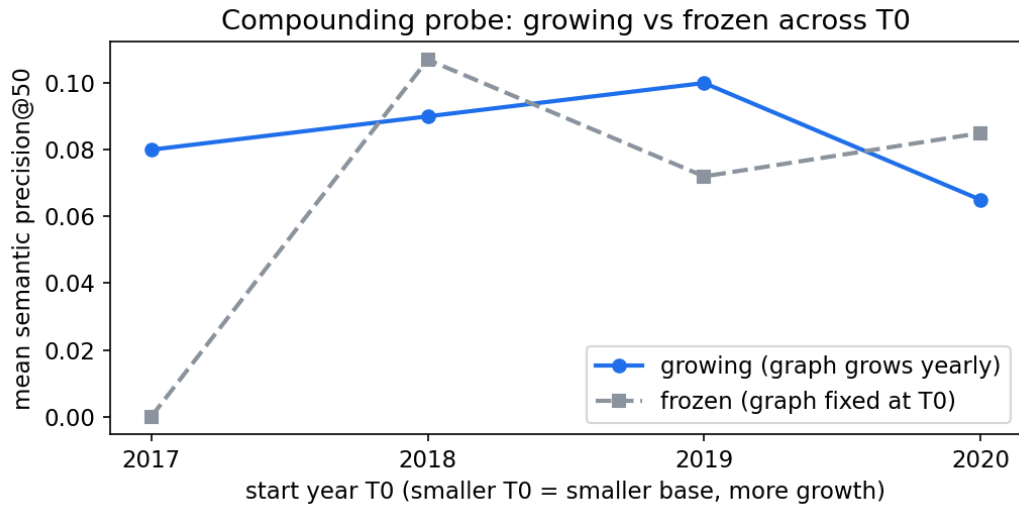


Figure 4. growing (graph grows each year) vs. frozen (graph fixed at T0) across start years T0.

5.3 Decisive paired tests

growing - (no-graph LLM) has a CI above 0 at every k (**significant**); growing - frozen has a consistently positive mean but a CI that still spans 0 (**not yet significant**).

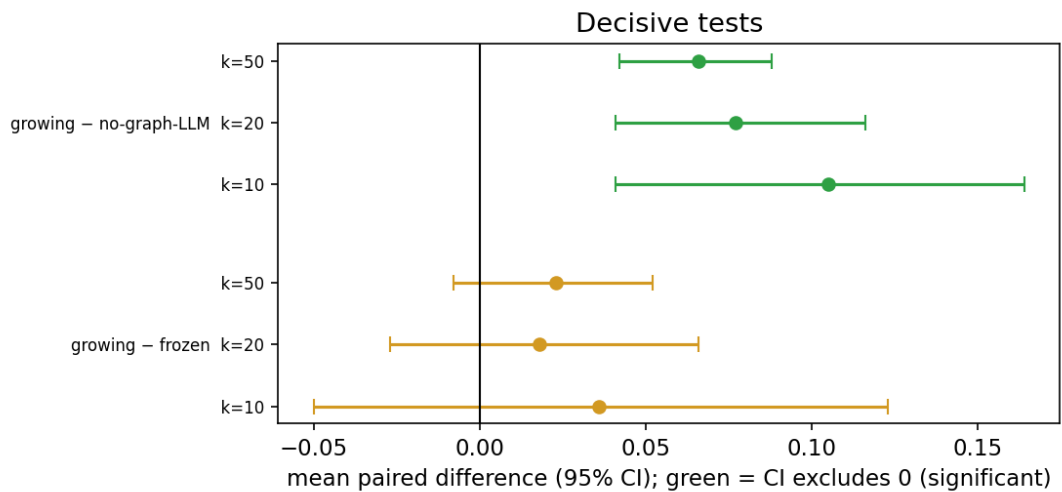


Figure 5. Paired mean differences with 95% CIs; green = CI excludes 0 (significant).

5.4 Edge-quality audit: transfer vs. usage

Classifying edges as genuine *transfer* (pretraining / representation reuse / multi-task / measured gain) vs. mere *usage* (a method simply applied to a task): only 20% of method->task edges are genuine transfer (78% usage), whereas 65% of task->task edges are transfer. Genuine-transfer edges are estimated at only ~23% of the graph.

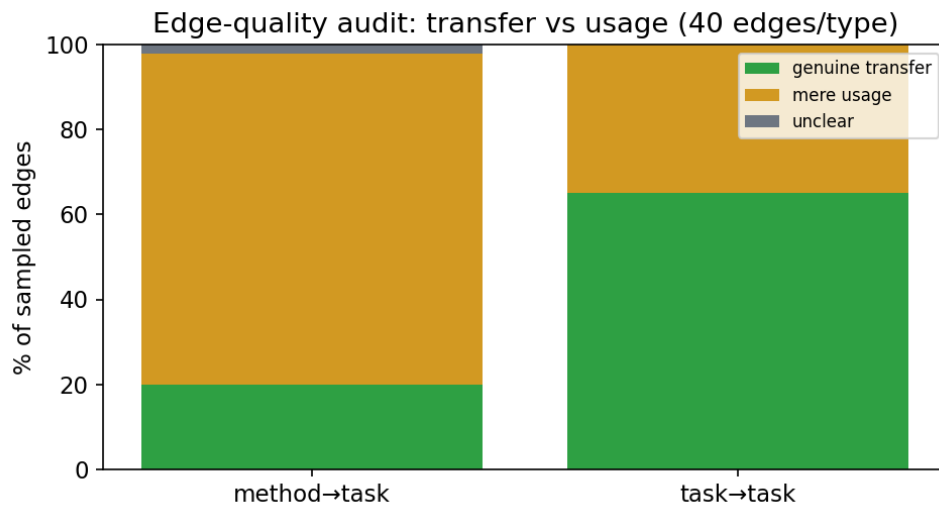


Figure 6. Edge-quality audit (40 sampled edges per type). The current graph is largely a method-task usage table; genuine transfer is the minority.

5.5 The extraction prompt is the key lever

A full-text vs. abstract A/B test shows that the main lever for thickening task->task is the **extraction prompt, not reading full text**. On the same abstracts, a transfer-targeted prompt raises the task->task share from 8% to 32%; adding full text dilutes the proportion (it contributes still more method->task edges).

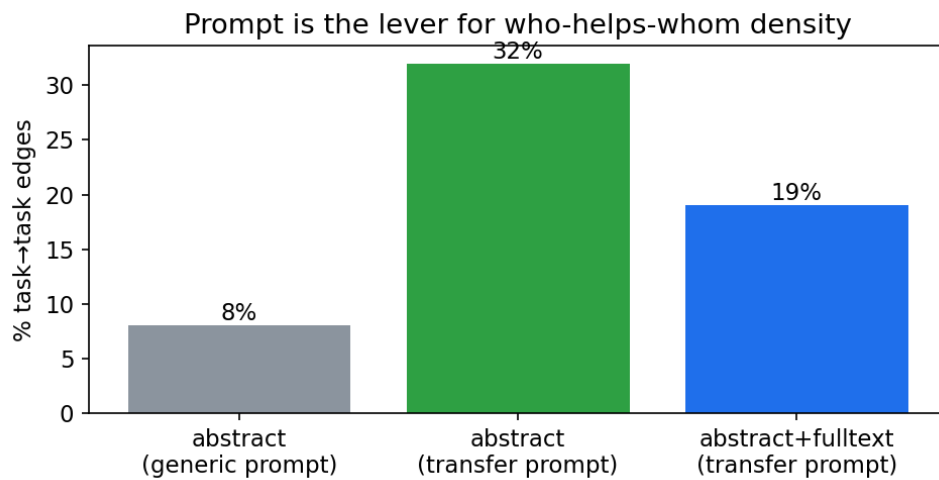


Figure 7. task->task edge share: generic prompt vs. transfer-targeted prompt vs. + full text.

6. Established / Open

Established (significant):

- Graph-structured navigation > no-graph LLM, > co-occurrence graph, >> random (paired CIs exclude 0).
- The gain is attributable to the graph, not to LLM leakage (the no-graph LLM is near-random).

Open:

- Compounding (growing > frozen) is directionally positive but not significant -- needs scale-up (more years / data points to tighten the CI).
- Edge quality: genuine transfer is only ~23% -- needs a transfer-vs-usage re-extraction.
- Scale and noise: 900-paper subsample, abstracts only, a single embedding model; values are small and noisy.

7. Full-Scale Outlook and Roadmap

Three highest-leverage steps (the rest is engineering scale-up):

- **Purifying re-extraction.** A prompt that distinguishes genuine transfer from usage (treating both source types equally); cheaply re-verify graph quality and whether “graph > baselines” survives purification.
- **Scale to resolve compounding.** ~3,000 papers plus more start years T0 to push the growing - frozen CI off zero; add full text to further thicken genuine transfer.
- **Full system.** Local 72B extraction (reproducible, leakage-controlled) + Neo4j / vector index + governance always on + multiple domains (brain -> retina -> cardiac) + a live auto-research closed-loop demo.

Stop-loss. If compounding remains non-significant after scale-up, fall back to the (already significant) “structure-driven navigation > no-graph LLM / co-occurrence / random” paper.

Implementation progress

Stage	Content	Status
P0	Data acquisition + graph scaffold	done (8,000 papers)
P1	Extraction validation	done
P2	Graph build + resolution + inference	done (886 papers)
P3	Graph navigation	done
P4-P6	Temporal evaluation + confidence intervals	done
P7-P8	Full-text A/B + edge-quality audit	done
--	Purifying re-extraction / scale-up / full system	to do